

The basics of decentralised applications

To understand decentralised applications, we must first grasp the concept of blockchain technology. Traditionally, applications rely on central servers to store and process data, leading to concerns about privacy, security, and single points of failure. Unlike centralised applications, decentralised applications operate on distributed networks that are not controlled by a single entity.

At the heart of DApps lies the blockchain, a distributed ledger that records transactions and interactions between participants. Smart contracts, which are self-executing agreements with predefined rules, automate processes and eliminate the need for intermediaries. This decentralised infrastructure ensures trust, transparency, and security.

Real-world DApp use cases

DApps are finding numerous applications across various industries. Here are some prominent real-world use cases of DApps.

Finance

Decentralised finance (DeFi) applications are gaining significant traction, offering users transparent and inclusive financial services. DApps like Compound, Aave, and Uniswap enable lending, borrowing, and decentralised trading without relying on traditional intermediaries.

Supply chain management

DApps provide an immutable and transparent record of supply chain activities. This enables businesses to track and verify the origin, authenticity, and movement of goods. Projects like VeChain and Origin Trail aim to optimise supply chain processes by leveraging the power of blockchain technology.

Gaming

DApps are disrupting the gaming industry by introducing ownership and interoperability of in-game assets. Games like Axie Infinity and Cryptokitties allow players to own, trade, and monetise virtual assets securely on the blockchain, creating a new paradigm for the gaming experience.

Governance

DApps are empowering decentralised governance models by providing transparent decision-making processes. Projects like Aragon and DAOstack enable token holders to participate in voting, proposal creation, and governance activities, allowing for decentralised and inclusive decision-making.

Advantages of decentralised applications

DApps offer a range of advantages.

Transparency and security

One of the key advantages of DApps is their transparency. Since all transactions are recorded on the blockchain, anyone can verify the integrity of the data. This transparency reduces the risk of fraud and promotes accountability. Moreover, the decentralised nature of DApps makes them resistant to hacking attempts. Unlike centralised databases, which are vulnerable to cyberattacks, the distributed nature of the blockchain adds an extra layer of security.

Elimination of intermediaries

In traditional applications, intermediaries such as banks, payment processors, or centralised exchanges facilitate transactions. These intermediaries often impose additional costs and delays. With DApps, intermediaries are largely eliminated. Smart contracts enable direct peer-to-peer transactions, cutting out unnecessary middlemen. This not only reduces costs but also accelerates the speed of transactions.

User control and ownership

With DApps, users gain greater control over their data and digital assets. Traditional applications often collect and monetise user data without consent. In contrast, DApps allow users to retain ownership of their data and determine how it is used. This shift in control empowers individuals and protects their privacy.

Global accessibility

Decentralised applications have the potential to bridge the global digital divide. Since they operate on the internet, DApps can be accessed by anyone with an internet connection. This opens opportunities for individuals in underserved regions to participate in the digital economy, access financial services, and engage in online transactions.

Challenges and limitations of decentralised applications

While decentralised applications offer numerous advantages, they also face certain challenges and limitations that must be addressed for widespread adoption.

Scalability

One of the primary challenges faced by DApps is scalability. Blockchain networks, especially those utilising proof-of-work consensus algorithms, have limited processing capacity. This results in slower transaction times and higher fees during periods of high demand. To overcome this challenge, various scaling solutions such as Layer 2 protocols, and improved consensus mechanisms are being developed.

User experience

The user experience in decentralised applications is often criticised for being complex and unintuitive. Interacting with DApps typically requires users to manage their private keys and understand the intricacies of blockchain technology. Improvements